David E. Bernal Neira

Website: secquoia.github.io/1-bernalde.html
Email: dbernaln@purdue.edu
LinkedIn: bernalde
GitHub: github.com/bernalde
ORCID: 0000-0002-8308-5016

Summary

Principal Investigator of the **S**ystems **E**ngineering via **C**lassical and **Qu**antum **O**ptimization for **I**ndustrial **A**pplications (SECQUOIA) research group and Mitchel Papanicolas Family/Del Ray Glass Company Assistant Professor in the Davidson School of Chemical Engineering at Purdue University. Co-director of the Center for Operations and Optimization in Process Systems (COOPS). Member, Inaugural Cohort of the Purdue College of Engineering Emerging Research Leaders.

Highlights

- **Research group:** 7 Ph.D. students and 2 postdoctoral associates currently advised at Purdue.
- **Publications:** 46 peer-reviewed journal articles, 22 conference proceedings, 2 book chapters and edited volumes, and 10 articles under review or in preparation.
- **Funding:** approximately $585,000 total across PI and Co-PI roles since joining Purdue in 2023.

Research Interests

- Optimization in Systems Engineering, with emphasis on energy and infrastructure systems, process intensification, and control.
- Quantum algorithms for combinatorial optimization, especially those applied to problems in chemical and process systems engineering.
- Design and evaluation of novel optimization and simulation algorithms targeting emerging computing hardware.
- Decision-making under uncertainty using optimization, machine learning, and artificial intelligence.
- Theory and algorithms for discrete-continuous nonlinear optimization, including software development.

Education

Carnegie Mellon University — Pittsburgh, PA, USA
Ph.D. in Chemical Engineering; Advisor: Prof. Ignacio E. Grossmann; GPA: 3.92/4.00 — Jan. 2017–May 2021

Universidad de los Andes — Bogotá, Colombia
B.S. in Physics; GPA: 4.62/5.00 — Jan. 2011–Mar. 2018
M.S. in Chemical Engineering; GPA: 4.73/5.00 — Aug. 2014–Oct. 2016
B.S. in Chemical Engineering with Honors (Cum Laude); GPA: 4.62/5.00 — Aug. 2010–Oct. 2014

Experience

Purdue University – Davidson School of Chemical Engineering — West Lafayette, IN, USA
Mitchel Papanicolas Family/Del Ray Glass Company Assistant Professor — Aug. 2023–Current

- Named to the Mitchel Papanicolas Family/Del Ray Glass Company Rising Star Professorship in 2026
- Principal investigator of the **S**ystems **E**ngineering via **C**lassical and **Qu**antum **O**ptimization for **I**ndustrial **A**pplications (SECQUOIA) research group

NASA – Universities Space Research Association (USRA) Mountain View, CA, USA

Visiting Scientist, USRA Research Institute of Advanced Computer Science (RIACS) Aug. 2023–Aug. 2024

Visiting Scientist, NASA Quantum and Artificial Intelligence Laboratory (QuAIL) Aug. 2023–Jun. 2024

Research Scientist, NASA QuAIL Jun. 2021–Jul. 2023

Associate Scientist, USRA RIACS Jun. 2021–Jul. 2023

- Research on optimization algorithms that leverage quantum computing for applications in science and engineering

Ph.D. Intern, Feynman Quantum Academy (NASA QuAIL and USRA RIACS) May 2019–Aug. 2019

- Development and implementation of a compiler for quantum annealing embeddings based on computational algebraic geometry and integer programming

Carnegie Mellon University – Tepper School of Business Pittsburgh, PA, USA

Adjunct Professor of Operations Management and Quantum Computing Aug. 2023–Dec. 2023

- Instructor for the Quantum Integer Programming and Machine Learning course offered jointly by Electrical and Computer Engineering and the Tepper School of Business

Carnegie Mellon University – Department of Chemical Engineering Pittsburgh, PA, USA

Visiting Research Scholar Jun. 2021–Aug. 2022

- Organized Grossmann Research Group meetings
- Maintained the MINLP and Generalized Disjunctive Programming library minlp.org

Ph.D. Student Jan. 2017–May 2021

- Developed algorithms for optimization in chemical, process, and energy systems
- Worked on methods for Mixed-Integer Nonlinear Programming and Generalized Disjunctive Programming
- Studied short-term quantum computing techniques for combinatorial optimization

Visiting Research Scholar May 2015–Aug. 2015

- Implemented heuristic algorithms for Mixed-Integer Nonlinear Programming in the solver DICOPT

ExxonMobil Engineering and Research Company Clinton, NJ, USA

Ph.D. Intern, Corporate Strategic Research Division May 2020–Aug. 2020

- Evaluated the potential of quantum computing for solving logistics-related optimization problems in the oil and gas industry

Ph.D. Intern, Process Technology Department May 2018–Aug. 2018

- Developed and deployed an optimal operation model for a combined heat and power plant with carbon capture

Universidad de los Andes – Department of Chemical Engineering Bogotá, Colombia

Graduate Teaching and Research Assistant Aug. 2014–Jul. 2016

- Conducted research in the Process and Products Design Group and the Process Optimization Group

Bayer Technology Services Leverkusen, Germany

Undergraduate Intern Feb. 2013–Jul. 2013

- Modeled dynamic flooding in distillation columns and simulated thermodynamic and electrolytic effects in HCl–water absorption systems for acid absorption columns

Grants and Sponsored Research

Total funding to date: approximately $585,000 across PI and Co-PI roles (∼$375,000 as PI; ∼$210,000 as Co-PI / Co-Investigator).

Principal Investigator – External and Industrial Funding Approx. $275,000

Decomposition Algorithms for Entropy Quantum Computers Apr. 2025–Apr. 2026

- Sponsor: Quantum Computing Incorporated (industrial partner).

- Total award amount: $103,467.

Quantum Machine Learning for Insurance Prediction — Nov. 2025–Nov. 2026

- Sponsor: Allstate (industrial partner).
- Total award amount: $248,110; $123,673 allocated to this investigator.

Efficient Mapping of Quadratic Integer Programming Problems into Qudit-Based Architectures — Aug. 2024–May 2025

- Sponsor: Quantum Collaborative Summit Seed Funds (Arizona State University center).
- Total award amount: $50,000.

Principal Investigator – Internal Seed Funding — Approx. $100,000

Efficient Algorithms for Optimization Using Analog and Gate-Based Quantum Computers — Jan. 2024–Jun. 2025

- Sponsor: Purdue College of Engineering and College of Science Seed Funds.
- Total award amount: $50,000.

Co-Principal Investigator / Co-Investigator – External Funding — Approx. $210,000

Hybrid Quantum Algorithms for Structured Optimization Problems in PSE — Dec. 2024–Dec. 2027

- Sponsor: National Science Foundation (NSF) Award 2430617.
- Total award amount: $550,000; $127,000 allocated to this investigator.
- Role: Co-Investigator.

Decomposition-Based Approaches for Practical Quantum Optimization — Jan. 2025–Dec. 2025

- Sponsor: NSF Center for Quantum Technologies Award 2224960.
- Total award amount: $37,000.
- Role: Co-Investigator; member allocation.

Enabling Quantum Computing Platform Access — Jan. 2023–Dec. 2023

- Sponsor: National Science Foundation (supplement to NSF Award 2132142).
- Total award amount: $48,572.
- Role: Co-Investigator; led supplement activities (base award PI: Zoltan Nagy).

Awards and Honors

Faculty Awards and Honors

- Mitchel Papanicolas Family/Del Ray Glass Company Rising Star Assistant Professorship — 2026
  Office of the Provost, Purdue University
- Inaugural Cohort – Purdue College of Engineering Emerging Research Leaders — 2026
  Purdue University College of Engineering
- Phillip C. Wankat Graduate Teaching Award in Chemical Engineering — 2025–2026
  Davidson School of Chemical Engineering, Purdue University
- Member – Young Academy of the Colombian Academy of Exact, Physical and Natural Sciences — 2025–2030
  Academia Colombiana de Ciencias Exactas, Físicas y Naturales (ACCEFYN)
- Invited Participant – Connections to Sustain Science in Latin America — 2025
  National Academies of Sciences, Engineering, and Medicine
- Best Paper Award – Assessing and advancing the potential of quantum computing: A NASA case study — 2024
  Future Generation Computer Systems Journal
- Invited Speaker – Arab-American Frontiers Symposium — 2023
  National Academies of Sciences, Engineering, and Medicine
- Best Talk – Quantum Computing Applications in Chemical and Biochemical Engineering Workshop — 2022
  American Institute of Chemical Engineers and Technical University of Denmark

Earlier Awards and Honors

- Finalist – AIChE CAST Directors' Student Presentation Award — 2020
  Computing and Systems Technology (CAST) Division – American Institute of Chemical Engineers
- Mark Dennis Karl Outstanding Teaching Assistant Award — 2019
  Department of Chemical Engineering – Carnegie Mellon University
- Graduated *Cum Laude* in Chemical Engineering — 2014
  Universidad de los Andes
- Valedictorian — 2009
  Gimnasio Británico
- First Place – National Physics Olympiad (Superior Level) — 2007
  Colombian Mathematics and Physics Olympiads

Fellowships and Scholarships

Faculty and Professional Fellowships

- Arab-American Frontiers Fellowship — 2024
  National Academies of Sciences, Engineering, and Medicine
- Feynman Quantum Academy Fellowship — 2019
  Research Institute of Advanced Computer Science – Universities Space Research Association

Graduate and Undergraduate Scholarships

- Fellowship for Ph.D. in Chemical Engineering — 2017
  Center for Advanced Process Decision-making (CAPD), Department of Chemical Engineering, Carnegie Mellon University
- Undergraduate Research Fellow in Astrophysics (SURF Cornell–UniAndes) — 2016
  Cornell University and Universidad de los Andes
- Fellowship for M.S. in Chemical Engineering — 2014
  Department of Chemical Engineering, Universidad de los Andes
- Young Engineers Scholarship for International Exchange at Otto-von-Guericke Universität — 2012
  German Academic Exchange Service (DAAD), COLCIENCIAS, and Universidad de los Andes
- Alberto Magno Scholarship for Academic Excellence — 2010–2014
  Universidad de los Andes

Teaching

Purdue University — West Lafayette, IN, USA
Course Instructor, Davidson School of Chemical Engineering — Fall 2023–Current

- **CHE 456 – Process Dynamics and Control** (required undergraduate course) — Fall 2023, 2024, 2025
- **CHE 597 – Data Science in Chemical Engineering** (graduate elective) — Spring 2025

Carnegie Mellon University — Pittsburgh, PA, USA
47-779 / 47-785 Quantum Integer Programming and Quantum Machine Learning – graduate course; **course originator** — Fall 2020–Fall 2023

- Adjunct Professor – Tepper School of Business and Department of Electrical and Computer Engineering — Fall 2023

– Invited Lecturer – Tepper School of Business and Department of Electrical and Computer Engineering Fall 2021, Fall 2022
– Course Instructor (Ph.D. student) – Tepper School of Business Fall 2020
– *Course launched at Tepper in Fall 2020; cross-listed with Electrical and Computer Engineering from Fall 2021 onward.*
– *Open course materials* (publicly available): QuIP 2020, QuIPML 2021, QuIPML 2022/23. Materials adapted for international short courses; see "Short Courses and Workshops Taught."

Graduate Teaching Assistant – Department of Chemical Engineering 2017–2018

– 06-421 Chemical Process Systems Design (undergraduate) Fall 2017, Fall 2018
  - *Mark Dennis Karl Outstanding Teaching Assistant Award*
– 06-720 Advanced Process Systems Engineering and 06-805 Disjunctive Programming (graduate) Spring 2018

Teaching-related Training and Development

– Future Faculty Program, Eberly Center, Carnegie Mellon University 2019–2022
– Teaching Effectiveness Colloquium – INFORMS 2020

Mentoring

Purdue University – Davidson School of Chemical Engineering West Lafayette, IN, USA
Current SECQUOIA Group: Ph.D. Candidates and Postdoctoral Associates 2023–Current

– Anja Hribljan – Ph.D. Candidate 2025–Current
– Mazhar Laljee – Ph.D. Candidate 2025–Current
– Sergey Gusev – Ph.D. Candidate 2024–Current
– Andrés Cabeza – Ph.D. Candidate 2024–Current
– Yirang Park – Ph.D. Candidate 2023–Current
– Anurag Ramesh – Ph.D. Candidate 2023–Current
– Albert Lee – Ph.D. Candidate 2022–Current
– Bruna Gabrielly De Moraes Araujo – Postdoctoral Associate 2026–Current
– Hanjing Xu – Postdoctoral Associate 2025–Current

Former Postdoctoral Associates

– Hamta Bardool – Postdoctoral Associate 2024–2025
– Carolina Tristán – Postdoctoral Associate 2024–2025
– Amandeep Singh Bhatia – Postdoctoral Associate 2023–2024
– Zedong Peng – Postdoctoral Associate 2023–2024

Visiting Faculty and Research Scholars

– Mohamed Taha Rouabah – Visiting Faculty Scholar, NASEM Arab-American Fellowship Spring 2024
– Chaolong Wang – Visiting Research Scholar Spring 2026
– Joao Victor Paim de Cerqueira Melo – Visiting Research Scholar Fall 2025
– Iago Leal de Freitas – Visiting Research Scholar Fall 2024
– Juan Sebastián Rodríguez – Visiting Research Scholar Spring 2024
– Andrés Cabeza – Visiting Research Scholar Fall 2023, Spring 2024
– Pedro Maciel Xavier – Visiting Research Scholar Fall 2023

Undergraduate Researchers

- *Visiting and Fellowship Scholars:* Alan Yi (SURF, Summer 2025); Daniel Anourou (SURF, Summer 2025); Mateo Huertas Marulanda (UREP-C, Spring 2025); André Lima Alambert (PONTES, Fall 2024).
- *Purdue Undergraduate Researchers:* Tarik Guler (Spring 2026); Alan Yi (Fall 2025–Spring 2026); Azain Khalid (Fall 2025); Sai Karthik (Spring 2025–Spring 2026); Akshay Mahajan (Spring 2025); AJ Collins (Spring 2025); Abigail Delaney, Lukas Peng, Dale Stager, Keegan Duffin, Dhruv Mendpara (Fall 2024); Benjamin Murray (Spring–Fall 2024); Sergio Barrios (Fall 2023).

NASA – Universities Space Research Association Mountain View, CA, USA
Feynman Quantum Academy 2021–2024

- Farshud Sorourifar – Ph.D., Chemical Engineering, Ohio State University
- Robin Brown – Ph.D., Computational and Mathematical Engineering, Stanford University
- Phillip Kerger – Ph.D., Applied Mathematics and Statistics, Johns Hopkins University
- Pratik Sathe – Ph.D., Physics, University of California, Los Angeles
- Dan Zhao – M.Sc., Computer Science, New York University
- Diana Chamaki – B.Sc., Physics, University of California, Berkeley
- Christopher Um – B.Sc., Physics, Cornell University

Externally Co-advised Master's Students

- Woosik Kim – M.Sc., Electrical and Computer Engineering, Purdue University 2025
- Pedro Maciel Xavier – M.Sc., Systems and Computing Engineering, Federal University of Rio de Janeiro, Brazil 2025
- Pedro da Silveira Carvalho Ripper – M.Sc., Electrical Engineering, Pontifical Catholic University of Rio de Janeiro, Brazil 2025

Earlier Mentees (Pre-Purdue)

Carnegie Mellon University – Chemical Engineering M.Sc. Student Research 2018–2020

- Yunshan Liu
- Haokun Yang

Carnegie Mellon University – Chemical Engineering Undergraduate Honors Research 2018–2020

- Felicity Gong
- Rahul Joglekar
- Saeed Syed
- Zhifei Yuliu

Universidad de los Andes – Chemical Engineering Undergraduate Theses 2015–2016

- Paola Cristancho
- Hugo Cuellar

Ph.D. Thesis Committee Member

- Gabriel Perez Schuster – Ph.D., Chemical Engineering, Purdue University. *Displacement Chromatography Methods for Producing Battery-Grade Lithium, Cobalt, Nickel, and Manganese from Black Mass and Mineral Concentrates.*
- Shrivatsa Korde – Ph.D., Chemical Engineering, Purdue University. *Recovering Active Pharmaceutical Ingredients from Unused Drug Products.*
- Sai Madhukiran Kompalli – Ph.D., Chemical Engineering, Purdue University. *A Decision Rule Framework for Interpretable Optimization.*

- Hao Chen – Ph.D., Chemical Engineering, Purdue University. *Machine Learning and Optimization: Efficiency, Explainability, and Effectiveness.*
- Hsuan-Hao Hsu – Ph.D., Chemical Engineering, Purdue University. *Computational Reaction Discovery Algorithms for Open-Shell and Ionic Organic Species.*
- Begum Yuksel – Ph.D., Chemical Engineering, Purdue University. *Atomistic Features Affecting Conductivity in Organic Mixed Ionic–Electronic Conductors.*
- Carolina Tristán – Ph.D., Chemical Engineering, Universidad de Santander, Cantabria, Spain. *Advancing Sustainability in the Water-Energy Nexus: Optimization of Reverse Electrodialysis Energy Recovery from Salinity Gradients.*
- Phillip Kerger – Ph.D., Applied Mathematics and Statistics, Johns Hopkins University. *Topics in Classical and Quantum Optimization: Complexity and Algorithms.*
- Guillermo Galán Iglesias – Ph.D., Chemical Engineering, Universidad de Salamanca, Spain. *Development of Tools for the Design of Processes at the Service of the Energy Transition.*
- Rodolfo Quintero – Ph.D., Industrial and Systems Engineering, Lehigh University. *Exact Penalization, Lagrangian Relaxation, and Applications to Quantum Computing.*

Student Awards and Honors

- Azain Khalid – NVIDIA'S Ecosystem Award — 2026
  iQuHack Hackathon, MIT
- Alan Yi – Fall Exposition Awardee — 2025
  Office of Undergraduate Research, Purdue University
- Sergey Gusev – Purdue Chemical Engineering Centennial Fellowship Award — 2025
  Davidson School of Chemical Engineering, Purdue University
- Andrés Cabeza – Best Student Paper Award — 2025
  American Institute of Chemical Engineers (AIChE), Process Development Division (PDD)
- Yirang Park – Travel Fellowship — 2025
  AIChE, Pharmaceutical Discovery, Development, and Manufacturing (PD2M) Forum
- Yirang Park – BioTrain Fellowship — 2025
  Young Institute for Pharmaceutical Manufacturing, Purdue University
- Andrés Cabeza – Best presentation at the AIChE LatinX Virtual Meeting — 2023
  American Institute of Chemical Engineers, LatinX Division
- Haokun Yang – Best submission in the "Refining and Petrochemical Plant Modeling and Operations Improvement" session: "Integration of Crude-Oil Scheduling and Refinery Planning by Lagrangean Decomposition Approach" — 2018
  AIChE Annual Meeting

Journal Publications

[J1] W. M. Brown, A. Ramesh, T. Lubinski, T. Nguyen, and **D. E. Bernal**, "Multi-gpu quantum circuit simulation and the impact of network performance", *Computer Physics Communications*, vol. 324, p. 110 126, 2026, ISSN: 0010-4655. DOI: 10.1016/j.cpc.2026.110126.

[J0] T. Koch, **D. E. Bernal**, Y. Chen, G. Cortiana, D. J. Egger, R. Heese, N. N. Hegade, A. G. Cadavid, R. Huang, T. Itoko, T. Kleinert, P. M. Xavier, N. Mohseni, J. A. Montanez-Barrera, K. Nakano, G. Nannicini, C. O'Meara, J. Pauckert, M. Proissl, A. Ramesh, M. Schicker, N. Shimada, M. Takeori, V. Valls, D. V. Bulck, S. Woerner, and C. Zoufal, "The Quantum Optimization Benchmarking Library", *Nature Computational Science*, vol. 6, pp. 653–671, 2026. DOI: 10.1038/s43588-026-00991-1.

[J0] J. A. Montanez-Barrera, Y. Ji, M. R. von Spakovsky, **D. E. Bernal**, and K. Michielsen, "Optimizing QAOA circuit transpilation with parity twine and SWAP network encodings", *Physical Review Applied*, 2026, Accepted for publication. DOI: 10.1103/sjqb-b1h4.

[J2] N. E. Belaloui, A. Tounsi, R. A. Khamadja, M. M. Louamri, A. Benslama, **D. E. Bernal**, and M. T. Rouabah, "Ground State Energy Estimation on Current Quantum Hardware Through The Variational Quantum Eigensolver: A Comprehensive Study", *Journal of Chemical Theory and Computation*, vol. 21, no. 14, pp. 6777–6792, 2025. DOI: 10.1021/acs.jctc.4c01657.

[J3] **D. E. Bernal**, R. Brown, P. Sathe, F. Wudarski, M. Pavone, E. G. Rieffel, and D. Venturelli, "Benchmarking the Operation of Quantum Heuristics and Ising Machines: Scoring Parameter Setting Strategies on Optimization Applications", *Quantum Machine Intelligence*, vol. 7, no. 86, 2025. DOI: 10.1007/s42484-025-00311-2.

[J4] **D. E. Bernal**, R. Herrman, M. Mohammadisiahroudi, G. Nannicini, R. Shaydulin, T. Terlaky, and S. Woerner, "What Can Quantum Computers Do for You?", *ORMS Today*, 2025. DOI: 10.1287/orms.2025.02.16.

[J5] A. F. Cabeza, **D. E. Bernal**, and Á. Orjuela, "Intensified production of butyl citrates from a calcium citrate salt via solid-liquid reaction", *Chemical Engineering Journal*, p. 164 032, 2025, ISSN: 1385-8947. DOI: 10.1016/j.cej.2025.164032.

[J6] A. Chatterjee, S. Rappaport, A. Giri, S. Johri, T. Proctor, **D. E. Bernal**, P. Sathe, and T. Lubinski, "A Comprehensive Cross-Model Framework for Benchmarking the Performance of Quantum Hamiltonian Simulations", *IEEE Transactions on Quantum Engineering*, vol. 6, pp. 1–26, 2025. DOI: 10.1109/TQE.2025.3558090.

[J7] M. Dupont, B. Sundar, B. Evert, **D. E. Bernal**, Z. Peng, S. Jeffrey, and M. J. Hodson, "Benchmarking quantum optimization for the maximum-cut problem on a superconducting quantum computer", *Phys. Rev. Appl.*, vol. 23, p. 014 045, 1 2025. DOI: 10.1103/PhysRevApplied.23.014045.

[J8] S. Dutta, I. Leal de Freitas, P. Maciel Xavier, C. Miceli de Farias, and **D. E. Bernal**, "Federated Learning in Chemical Engineering: A Tutorial on a Framework for Privacy-Preserving Collaboration across Distributed Data Sources", *Industrial & Engineering Chemistry Research*, 2025. DOI: 10.1021/acs.iecr.4c03805.

[J9] E. J. Gustafson, J. Tiihonen, D. Chamaki, F. Sorourifar, J. W. Mullinax, A. C. Y. Li, F. B. Maciejewski, N. P. D. Sawaya, J. T. Krogel, **D. E. Bernal**, and N. M. Tubman, "Surrogate optimization of variational quantum circuits", *Proceedings of the National Academy of Sciences*, vol. 122, no. 36, 2025. DOI: 10.1073/pnas.2408530122.

[J10] J. Kronqvist, **D. E. Bernal**, and I. E. Grossmann, "50 years of mixed-integer nonlinear and disjunctive programming", *European Journal of Operational Research*, 2025. DOI: 10.1016/j.ejor.2025.07.016.

[J11] A. J. Lee and **D. E. Bernal**, "Mixed-integer reaggregated hull reformulation of special structured generalized linear disjunctive programs", *Industrial & Engineering Chemistry Research*, 2025. DOI: 10.1021/acs.iecr.5c03172.

[J12] D. Ovalle, D. A. Liñán, A. Lee, J. M. Gómez, L. Ricardez-Sandoval, I. E. Grossmann, and **D. E. Bernal**, "Logic-Based Discrete-Steepest Descent: A Solution Method for Process Synthesis Generalized Disjunctive Programs", *Computers & Chemical Engineering*, p. 108 993, 2025. DOI: 10.1016/j.compchemeng.2024.108993.

[J13] F. Sorourifar, M. T. Rouabah, N. E. Belaloui, M. M. Louamri, D. Chamaki, E. J. Gustafson, N. M. Tubman, J. A. Paulson, and **D. E. Bernal**, "Toward efficient quantum computation of molecular ground-state energies", *AIChE Journal*, e18887, 2025. DOI: 10.1002/aic.18887.

[J14] **D. E. Bernal** and I. E. Grossmann, "Convex mixed-integer nonlinear programs derived from generalized disjunctive programming using cones", *Computational Optimization and Applications*, pp. 1–62, 2024. DOI: 10.1007/s10589-024-00557-9.

[J15] **D. E. Bernal**, C. D. Laird, L. R. Lueg, S. M. Harwood, D. Trenev, and D. Venturelli, "Utilizing modern computer architectures to solve mathematical optimization problems: A survey", *Computers & Chemical Engineering*, p. 108 627, 2024. DOI: 10.1016/j.compchemeng.2024.108627.

[J16] A. S. Bhatia and **D. E. Bernal**, "Federated learning with tensor networks: a quantum AI framework for healthcare", *Machine Learning: Science and Technology*, vol. 5, no. 4, p. 045 035, 2024. DOI: 10.1088/2632-2153/ad8c11.

[J17] R. Brown, **D. E. Bernal**, D. Venturelli, and M. Pavone, "A copositive framework for analysis of hybrid Ising-classical algorithms", *SIAM Journal on Optimization*, vol. 34, no. 2, pp. 1455–1489, 2024. DOI: 10.1137/22M1514581.

[J18] T. Lubinski, C. Coffrin, C. McGeoch, P. Sathe, J. Apanavicius, and **D. E. Bernal**, "Optimization Applications as Quantum Performance Benchmarks", *ACM Transactions on Quantum Computing*, 2024. DOI: 10.1145/3678184.

[J19] E. G. Rieffel, A. A. Asanjan, M. S. Alam, N. Anand, **D. E. Bernal**, S. Block, L. T. Brady, S. Cotton, Z. G. Izquierdo, S. Grabbe, *et al.*, "Assessing and advancing the potential of quantum computing: A NASA case study", *Future Generation Computer Systems*, vol. 160, pp. 598–618, 2024, **Best paper award**. DOI: 10.1016/j.future.2024.06.012.

[J20] N. P. Sawaya, D. Marti-Dafcik, Y. Ho, D. P. Tabor, **D. E. Bernal**, A. B. Magann, S. Premaratne, P. Dubey, A. Matsuura, N. Bishop, W. A. d. Jong, S. Benjamin, O. Parekh, N. Tubman, K. Klymko, and D. Camps, "HamLib: A library of Hamiltonians for benchmarking quantum algorithms and hardware", *Quantum*, vol. 8, p. 1559, Dec. 2024, ISSN: 2521-327X. DOI: 10.22331/q-2024-12-11-1559.

[J21] J. Wang, Z. Peng, R. Hughes, D. Bhattacharyya, **D. E. Bernal**, and A. W. Dowling, "Measure This, Not That: Optimizing the Cost and Model-Based Information Content of Measurements", *Computers & Chemical Engineering*, vol. 189, p. 108 786, 2024. DOI: 10.1016/j.compchemeng.2024.108786.

[J22] P. Kerger, **D. E. Bernal**, Z. Gonzalez Izquierdo, and E. G. Rieffel, "Mind the $\tilde{O}$: Asymptotically Better, but Still Impractical, Quantum Distributed Algorithms", *Algorithms*, vol. 16, no. 7, 2023, ISSN: 1999-4893. DOI: 10.3390/a16070332.

[J23] L. Su, **D. E. Bernal**, I. E. Grossmann, and L. Tang, "Modeling for integrated refinery planning with crude-oil scheduling", *Chemical Engineering Research and Design*, vol. 192, pp. 141–157, 2023. DOI: 10.1016/j.cherd.2023.02.008.

[J24] **D. E. Bernal**, "Coherent simulation with thousands of qubits", *Nature Physics*, 2022. DOI: 10.1038/s41567-022-01772-z.

[J25] **D. E. Bernal**, A. Ajagekar, S. M. Harwood, S. T. Stober, D. Trenev, and F. You, "Perspectives of Quantum Computing for Chemical Engineering", *AIChE Journal*, e17651, 2022. DOI: 10.1002/aic.17651.

[J26] **D. E. Bernal**, Z. Peng, J. Kronqvist, and I. E. Grossmann, "Alternative regularizations for Outer-Approximation algorithms for convex MINLP", *Journal of Global Optimization*, pp. 1–36, 2022. DOI: 10.1007/s10898-022-01178-4.

[J27] R. Quintero, **D. E. Bernal**, T. Terlaky, and L. F. Zuluaga, "Characterization of QUBO reformulations for the maximum k-colorable subgraph problem", *Quantum Information Processing*, vol. 21, no. 3, pp. 1–36, 2022. DOI: 10.1007/s11128-022-03421-z.

[J28] Q. Chen, E. S. Johnson, **D. E. Bernal**, R. Valentin, S. Kale, J. Bates, J. D. Siirola, and I. E. Grossmann, "Pyomo.GDP: an ecosystem for logic based modeling and optimization development", *Optimization and Engineering*, pp. 1–36, 2021. DOI: 10.1007/s11081-021-09601-7.

[J29] S. Harwood, C. Gambella, D. Trenev, A. Simonetto, **D. Bernal**, and D. Greenberg, "Formulating and Solving Routing Problems on Quantum Computers", *IEEE Transactions on Quantum Engineering*, vol. 2, pp. 1–17, 2021. DOI: 10.1109/TQE.2021.3049230.

[J30] D. A. Liñán, **D. E. Bernal**, J. M. Gómez, and L. A. Ricardez-Sandoval, "Optimal synthesis and design of catalytic distillation columns: A rate-based modeling approach", *Chemical Engineering Science*, vol. 231, p. 116 294, 2021. DOI: 10.1016/j.ces.2020.116294.

[J31] H. A. Pedrozo, S. R. Reartes, **D. E. Bernal**, A. Vecchietti, M. S. Díaz, and I. E. Grossmann, "Hybrid model generation for superstructure optimization with Generalized Disjunctive Programming", *Computers & Chemical Engineering*, vol. 154, p. 107 473, 2021. DOI: 10.1016/j.compchemeng.2021.107473.

[J32] **D. E. Bernal**, S. Vigerske, F. Trespalacios, and I. E. Grossmann, "Improving the performance of DICOPT in convex MINLP problems using a feasibility pump", *Optimization Methods and Software*, vol. 35, no. 1, pp. 171–190, 2020. DOI: 10.1080/10556788.2019.1641498.

[J33] T. J. Ikonen, H. Mostafaei, Y. Ye, **D. E. Bernal**, I. E. Grossmann, and I. Harjunkoski, "Large-scale selective maintenance optimization using bathtub-shaped failure rates", *Computers & Chemical Engineering*, vol. 139, p. 106 876, 2020. DOI: 10.1016/j.compchemeng.2020.106876.

[J34] J. Kronqvist, **D. E. Bernal**, and I. E. Grossmann, "Using regularization and second order information in outer approximation for convex MINLP", *Mathematical Programming*, vol. 180, no. 1, pp. 285–310, 2020. DOI: 10.1007/s10107-018-1356-3.

[J35] C. Li, **D. E. Bernal**, K. C. Furman, M. A. Duran, and I. E. Grossmann, "Sample average approximation for stochastic nonconvex mixed integer nonlinear programming via outer-approximation", *Optimization and Engineering*, pp. 1–29, 2020. DOI: 10.1007/s11081-020-09563-2.

[J36] D. A. Liñán, **D. E. Bernal**, L. A. Ricardez-Sandoval, and J. M. Gómez, "Optimal design of superstructures for placing units and streams with multiple and ordered available locations. Part I: A new mathematical framework", *Computers & Chemical Engineering*, p. 106 794, 2020. DOI: 10.1016/j.compchemeng.2020.106794.

[J37] D. A. Liñán, **D. E. Bernal**, L. A. Ricardez-Sandoval, and J. M. Gómez, "Optimal design of superstructures for placing units and streams with multiple and ordered available locations. Part II: Rigorous design of catalytic distillation columns", *Computers & Chemical Engineering*, p. 106 845, 2020. DOI: 10.1016/j.compchemeng.2020.106845.

[J38] H. Yang, **D. E. Bernal**, R. E. Franzoi, F. G. Engineer, K. Kwon, S. Lee, and I. E. Grossmann, "Integration of Crude-Oil Scheduling and Refinery Planning by Lagrangean Decomposition", *Computers & Chemical Engineering*, p. 106 812, 2020. DOI: 10.1016/j.compchemeng.2020.106812.

[J39] J. Kronqvist, **D. E. Bernal**, A. Lundell, and I. E. Grossmann, "A review and comparison of solvers for convex MINLP", *Optimization and Engineering*, vol. 20, no. 2, pp. 397–455, 2019. DOI: 10.1007/s11081-018-9411-8.

[J40] J. Kronqvist, **D. E. Bernal**, A. Lundell, and T. Westerlund, "A center-cut algorithm for quickly obtaining feasible solutions and solving convex MINLP problems", *Computers & Chemical Engineering*, vol. 122, pp. 105–113, 2019. DOI: 10.1016/j.compchemeng.2018.06.019.

[J41] C. L. Lara, **D. E. Bernal**, C. Li, and I. E. Grossmann, "Global optimization algorithm for multi-period design and planning of centralized and distributed manufacturing networks", *Computers & Chemical Engineering*, vol. 127, pp. 295–310, 2019. DOI: 10.1016/j.compchemeng.2019.05.022.

[J42] **D. E. Bernal**, C. Carrillo-Diaz, J. M. Gómez, and L. A. Ricardez-Sandoval, "Simultaneous design and control of catalytic distillation columns using comprehensive rigorous dynamic models", *Industrial & Engineering Chemistry Research*, vol. 57, no. 7, pp. 2587–2608, 2018. DOI: 10.1021/acs.iecr.7b04205.

[J43] L. Su, L. Tang, **D. E. Bernal**, and I. E. Grossmann, "Improved quadratic cuts for convex mixed-integer nonlinear programs", *Computers & Chemical Engineering*, vol. 109, pp. 77–95, 2018, ISSN: 0098-1354. DOI: 10.1016/j.compchemeng.2017.10.011.

[J44] L. Leisman, M. P. Haynes, S. Janowiecki, G. Hallenbeck, G. Józsa, R. Giovanelli, E. A. Adams, **D. E. Bernal**, J. M. Cannon, W. F. Janesh, *et al.*, "(Almost) Dark Galaxies in the ALFALFA Survey: Isolated H i-bearing Ultra-diffuse Galaxies", *The Astrophysical Journal*, vol. 842, no. 2, p. 133, 2017. DOI: 10.3847/1538-4357/aa7575.

Book Chapters and Editions

[B1] V. Aggarwal, **D. E. Bernal**, and D. Yu, Eds., *Frontiers in Industrial Engineering. Purdue Quantum AI, Fifth Gavriel Salvendy International Symposium, PQAI 2025, West Lafayette, IN, USA, October 23–24, 2025, Proceedings*, vol. 2724, Communications in Computer and Information Science, Springer Singapore, Jul. 2026, ISBN: 978-981-95-7828-3. DOI: 10.1007/978-981-95-7829-0.

[B2] **D. E. Bernal**, F. Gómez-Castro, E. A. del-Rio-Chanona, and V. Rico-Ramírez, "Future insights for optimization in chemical engineering", in *Optimization in Chemical Engineering: Deterministic, Meta-Heuristic and Data-Driven Techniques*, ser. De Gruyter Textbook, F. Gómez-Castro and V. Rico-Ramírez, Eds., De Gruyter, 2025, ISBN: 9783111383620.

Conference Proceedings

[C1] S. Dutta, N. Innan, S. B. Yahia, M. Shafique, and **D. E. Bernal**, "MQFL-FHE: Multimodal Quantum Federated Learning Framework with Fully Homomorphic Encryption", in *2025 International Joint Conference on Neural Networks (IJCNN)*, 2025, pp. 1–10. DOI: 10.1109/IJCNN64981.2025.11228914.

[C2] I. L. de Freitas, J. V. P. de Cerqueira Melo, and **D. E. Bernal**, "Quantum-Inspired Tensor Network Methods for Quadratic Unconstrained Binary Optimization", in *NeurIPS 2025 Workshop ScaleOPT: GPU-Accelerated and Scalable Optimization*, 2025.

[C3] S. Niu, E. Kökcü, S. Johri, A. Ramesh, A. Chatterjee, **D. E. Bernal**, D. Camps, and T. Lubinski, "A Practical Framework for Assessing the Performance of Observable Estimation in Quantum Simulation", in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, vol. 01, 2025, pp. 375–386. DOI: 10.1109/QCE65121.2025.00050.

[C4] Z. Peng, D. de Roux, and **D. E. Bernal**, “Hybrid quantum branch-and-bound method for quadratic unconstrained binary optimization”, in *Quantum Informatics, Computing & Technology 2025 (QC–HORIZON) 2025*, ser. Computer Science Research Notes, vol. 2, 2025, pp. 87–96. DOI: 10.24132/CSRN.2025-A67.

[C5] R. A. Brown, D. Venturelli, M. Pavone, and **D. E. Bernal**, “Accelerating continuous variable coherent ising machines via momentum”, in *International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research*, **(CPAIOR2024)**, Springer, 2024, pp. 109–126. DOI: 10.1007/978-3-031-60597-0_8.

[C6] A. F. Cabeza, A. Orjuela, and **D. E. Bernal**, “A Novel Cost-Efficient Tributyl Citrate Production Process”, in *Proceedings of the 10th International Conference on Foundations of Computer Aided Process Design (FOCAPD 2024)*, ser. Systems & Controls Transactions, vol. 3, **(FOCAPD2024)**, PSE Press: Hamilton, 2024, pp. 121–128, ISBN: 978-1-7779403-2-4. DOI: 10.69997/sct.122277.

[C7] A. F. Cabeza, A. Orjuela, and **D. E. Bernal**, “Analysis of Calcium Citrate Salts as Raw Material for Tributyl Citrate Bio-Plasticizer Production: Kinetic Modeling, Process Simulation, and Optimization”, in *Computer Aided Chemical Engineering*, vol. 53, **(PSE2024/ESCAPE34)**, Elsevier, 2024, pp. 955–960. DOI: 10.1016/B978-0-443-28824-1.50160-5.

[C8] S. Dutta, P. P. Karanth, P. M. Xavier, I. L. de Freitas, N. Innan, S. B. B. Yahia, M. Shafique, and **D. E. Bernal**, “Federated Learning with Quantum Computing and Fully Homomorphic Encryption: A Novel Computing Paradigm Shift in Privacy-Preserving ML”, in *NeurIPS 2024 Workshop Machine Learning with new Compute Paradigms*, 2024.

[C9] F. B. Maciejewski, B. G. Bach, M. Dupont, P. A. Lott, B. Sundar, **D. E. Bernal**, I. Safro, and D. Venturelli, “A Multilevel Approach for Solving Large-Scale QUBO Problems with Noisy Hybrid Quantum Approximate Optimization”, in *2024 IEEE High Performance Extreme Computing Conference (HPEC)*, 2024, pp. 1–10. DOI: 10.1109/HPEC62836.2024.10938438.

[C10] Z. Peng, K. Cao, K. C. Furman, C. Li, I. E. Grossmann, and **D. E. Bernal**, “A Convexification-Based Outer-Approximation Method for Convex and Nonconvex MINLP”, in *Computer Aided Chemical Engineering*, **(PSE2024/ESCAPE34)**, Elsevier, vol. 53, 2024. DOI: 10.1016/B978-0-443-28824-1.50536-6.

[C11] Z. Peng, A. Lee, and **D. E. Bernal**, “Addressing Discrete Dynamic Optimization via a Logic-Based Discrete-Steepest Descent Algorithm”, in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, 2024, pp. 1664–1669. DOI: 10.1109/CDC56724.2024.10886477.

[C12] F. Sorourifar, D. Chamaki, N. M. Tubman, J. Paulson, and **D. E. Bernal**, “Bayesian Optimization Priors for Efficient Variational Quantum Algorithms”, in *Computer Aided Chemical Engineering*, vol. 53, **(PSE2024/ESCAPE34)**, Elsevier, 2024, pp. 3379–3384. DOI: 10.1016/B978-0-443-28824-1.50564-0.

[C13] C. Tristán, M. Fallanza, R. Ibáñez, I. E. Grossmann, and **D. E. Bernal**, “Designing Reverse Electrodialysis Process for Salinity Gradient Power Generation via Disjunctive Programming”, in *Proceedings of the 10th International Conference on Foundations of Computer Aided Process Design (FOCAPD 2024)*, ser. Systems & Controls Transactions, vol. 3, **(FOCAPD2024)**, PSE Press: Hamilton, 2024, pp. 904–911, ISBN: 978-1-7779403-2-4. DOI: 10.69997/sct.126079.

[C14] C. Tristán, M. Fallanza, R. Ibáñez, I. E. Grossmann, and **D. E. Bernal**, “Global Optimization via Quadratic Disjunctive Programming for Water Networks Design with Energy Recovery”, in *Computer Aided Chemical Engineering*, vol. 53, **(PSE2024/ESCAPE34)**, Elsevier, 2024, pp. 2161–2166. DOI: 10.1016/B978-0-443-28824-1.50361-6.

[C15] P. M. Xavier, P. Ripper, J. Pulsipher, J. D. Garcia, N. Maculan, and **D. E. Bernal**, “Disjunctive Programming meets QUBO”, in *Computer Aided Chemical Engineering*, vol. 53, **(PSE2024/ESCAPE34)**, Elsevier, 2024, pp. 3433–3438. DOI: 10.1016/B978-0-443-28824-1.50573-1.

[C16] P. A. Kerger, **D. E. Bernal**, Z. G. Izquierdo, and E. G. Rieffel, “Quantum Distributed Algorithms for Approximate Steiner Trees and Directed Minimum Spanning Trees”, in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, vol. 01, 2023, pp. 1249–1259. DOI: 10.1109/QCE57702.2023.00141.

[C17] N. P. Sawaya, D. Marti-Dafcik, Y. Ho, D. P. Tabor, **D. E. Bernal**, A. B. Magann, S. Premaratne, P. Dubey, A. Matsuura, N. Bishop, W. A. De Jong, S. Benjamin, O. D. Parekh, N. M. Tubman, K. Klymko, and D. Camps, “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware”, in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, vol. 02, 2023, pp. 389–390. DOI: 10.1109/QCE57702.2023.10296.

[C18] **D. E. Bernal**, Y. Liu, M. L. Bynum, C. D. Laird, J. D. Siirola, and I. E. Grossmann, "Advances in Generalized Disjunctive and Mixed-Integer Nonlinear Programming Algorithms and Software for Superstructure Optimization", in *Computer Aided Chemical Engineering*, vol. 49, **(PSE2021+)**, Elsevier, 2022, pp. 1285–1290. DOI: 10.1016/B978-0-323-85159-6.50214-1.

[C19] **D. E. Bernal**, D. Ovalle, D. A. Liñán, L. A. Ricardez-Sandoval, J. M. Gómez, and I. E. Grossmann, "Process Superstructure Optimization through Discrete Steepest Descent Optimization: a GDP Analysis and Applications in Process Intensification", in *Computer Aided Chemical Engineering*, vol. 49, **(PSE2021+)**, Elsevier, 2022, pp. 1279–1284. DOI: 10.1016/B978-0-323-85159-6.50213-X.

[C20] **D. E. Bernal**, K. E. Booth, R. Dridi, H. Alghassi, S. Tayur, and D. Venturelli, "Integer programming techniques for minor-embedding in quantum annealers", in *International Conference on Integration of Constraint Programming, Artificial Intelligence, and Operations Research*, **(CPAIOR2020)**, Springer, 2020, pp. 112–129. DOI: 10.1007/978-3-030-58942-4_8.

[C21] **D. E. Bernal**, Q. Chen, F. Gong, and I. E. Grossmann, "Mixed-Integer Nonlinear Decomposition Toolbox for Pyomo (MindtPy)", in *13th International Symposium on Process Systems Engineering (PSE 2018)*, ser. Computer Aided Chemical Engineering, vol. 44, **(PSE2018)**, Elsevier, 2018, pp. 895–900. DOI: 10.1016/B978-0-444-64241-7.50144-0.

[C22] L. Su, L. Tang, **D. E. Bernal**, I. E. Grossmann, and B. Wang, "Integrated scheduling of on-line blending and distribution of oil products in refinery operation", in *13th International Symposium on Process Systems Engineering (PSE 2018)*, ser. Computer Aided Chemical Engineering, vol. 44, **(PSE2018)**, Elsevier, 2018, pp. 1213–1218. DOI: 10.1016/B978-0-444-64241-7.50197-X.

Preprints and Articles Under Review

[S0] M. Guo, J. Jurado Diaz, A. Ramesh, C. J. Haupt, A. Baiardi, D. Athanasakos, M. E. Sahin, O. Wallis, G. Pennington, C. Arenz, S. Brandhofer, G. Korpas, I. Čepaitė, J. A. Montañez-Barrera, J. Marecek, D. Venturelli, S. Eidenbenz, **D. E. Bernal**, and D. J. Egger, "Setting angles in quantum approximate optimization at utility-scale", arXiv preprint arXiv:2606.05311. Available here, 2026.

[S1] Y. Park and **D. E. Bernal**, "Leveraging Classical and Quantum Computing for Process Systems Engineering Applications: Decomposition Algorithm with Ising Solvers for Efficient Discrete Landscape Exploration", Submitted for publication. Available here, 2026.

[S2] S. Gusev and **D. E. Bernal**, "Exact Hull Reformulation for Quadratically Constrained Generalized Disjunctive Programs", Submitted for publication. Available here, 2025.

[S5] J. A. Montanez-Barrera, K. Michielsen, and **D. E. Bernal**, "Evaluating the performance of quantum processing units at large width and depth", Submitted for publication. Available here, 2025.

[S6] D. de Roux, Z. Peng, and **D. E. Bernal**, "Spectral Outer-Approximation Algorithms for Binary Semidefinite Problems", Submitted for publication. Available here, 2025.

[S7] A. Tounsi, N. E. Belaloui, A. R. Khamadja, T. E. F. Lalaoui, M. M. Louamri, **D. E. Bernal**, and M. T. Rouabah, "Probing the Ground State of the Antiferromagnetic Heisenberg Model on the Kagome Lattice using Geometrically Informed Variational Quantum Eigensolver", Available here, 2025.

[S8] Z. Peng, K. Cao, K. Furman, C. Li, I. E. Grossmann, and **D. E. Bernal**, "Enhanced Outer-Approximation Methods for MINLP via Convexification and Bound Tightening", Under Review., 2024.

[S9] W. Chaimanowong, **D. E. Bernal**, and F. Cisternas, "Optimizing Product Influence of Shelf Display", Submitted for publication. Available here, 2023.

[S10] P. Maciel Xavier, P. Ripper, T. Andrade, J. Dias Garcia, N. Maculan, and **D. E. Bernal**, "QUBO.jl: A Julia Ecosystem for Quadratic Unconstrained Binary Optimization", Submitted for publication. Available here, 2023.

[S11] **D. E. Bernal**, S. Tayur, and D. Venturelli, "Quantum Integer Programming (QuIP) 47-779: Lecture Notes", Available here, 2020.

Software Products

- **MindtPy** (Developer): Open-source decomposition-based toolkit for Mixed-Integer Nonlinear Programming. Integrated as a core solver in the Pyomo algebraic modeling ecosystem (U.S. DOE / Sandia National Laboratories).
- **QUBO.jl** (Originator and lead developer): Julia library for reformulating, analyzing, manipulating, and solving mathematical programs as Quadratic Unconstrained Binary Optimization problems. Part of the JuliaQUBO organization and the JuMP mathematical-programming ecosystem.
- **GDPLib** (Maintainer): Open-source library of Generalized Disjunctive Programming models implemented in Pyomo; used as a standard benchmark library in MINLP and GDP solver development.
- **Stochastic-Benchmark** (Developer): Open-source benchmarking framework for stochastic optimization solvers; developed at USRA-RIACS and used by the NASA QuAIL laboratory.
- **GDPOpt** (Contributor): Open-source solver for Generalized Disjunctive Programming in Pyomo.
- **PySA** (Contributor): Open-source Fast Simulated Annealing in native Python; released by NASA.
- **DICOPT** (Contributor): Commercial solver for Mixed-Integer Nonlinear Programming, distributed with GAMS.

Short Courses and Workshops Taught

- Invited Workshop: Introduction to Quantum Computing for Chemical Engineers – American Institute for Chemical Engineers Annual Meeting, Boston, Massachusetts, USA Nov. 2025
- Invited Short Course: Data Science and Optimization in Chemical Engineering – Tecnológico de Monterrey, Monterrey, México Jan. 2025
- Invited Lecture: Reformulations and Decomposition for Quantum Discrete Optimization: applications in optimal power flow – 6th Grid Science Winter School – Los Alamos National Laboratory, Santa Fe, NM, USA Jan. 2025
- Invited Summer School: Optimization in Biorefineries – Universidad Nacional, Bogotá, Colombia Jul. 2024
- Workshop: Practical workshop on Quantum Computing for Optimization for Process Systems Engineering – PSE/ESCAPE34, Florence, Italy Jun. 2024
- Invited Master Class: Quantum Computing for CP, AI, and OR, and vice-versa: Quantum-Classical Hybrid Methods for Optimization – CPAIOR, Uppsala, Sweden May 2024
- Invited Workshop: Quantum Integer Programming – Key Laboratory of Data Analytics and Optimization for Smart Industry, Northeastern University, Shenyang, China Apr. 2024
- Invited Short Course: Quantum Computing in Optimization – Electrical Engineering and Computer Science, Khalifa University, Abu Dhabi, United Arab Emirates Mar. 2024
- Invited Short Course Lectures: 2023 Gene Golub SIAM Summer School on Quantum Computing and Optimization – Lehigh University, Bethlehem, PA, USA Aug. 2023
- Invited Lecture: Discrete nonlinear optimization: Modeling and solutions via novel hardware and decomposition algorithms – 5th Grid Science Winter School – Los Alamos National Laboratory, Santa Fe, NM, USA Jan. 2023
- Short Course: Mixed-Integer and Disjunctive Optimization Theory, Software, and Algorithms – Institute of Industrial & Systems Engineering, Northeastern University, Shenyang, China Apr. 2019
- Software Workshop: Mathematical Programming in Python/Pyomo – Instituto de Desarrollo y Diseño INGAR CONICET–Universidad Tecnológica Nacional, Santa Fé, Argentina Aug. 2018

Invited Seminars and Lectures

Plenary, Keynote, and Distinguished Lectures

- Invited Keynote: "Perspectives on Quantum Computing for Chemical Engineering" – 17th AIChE Midwest Regional Conference, Chicago, IL, USA Apr. 2025
- Plenary Moderator and Panelist: "Future of Quantum Computing in Optimization" – 2022 CORS/INFORMS International Conference, Vancouver, Canada Jun. 2022

- Plenary: “Modern Computational Approaches to Nonlinear Discrete Optimization and Process Systems Engineering” (in Spanish) – Argentinian Symposium on Industrial Computing and Operations Research, Argentina Video Aug. 2021

Invited Seminars at US Universities

- “Is this quantum computer useful? Let’s solve optimization with it! Evaluating the performance of quantum processing units at large width and depth” – Industrial and Systems Engineering Department Seminar, University of Tennessee, Knoxville, TN Apr. 2026
- “Perspectives on Quantum Computing for Chemical Engineering” – Chemical and Biomolecular Engineering Department Seminar, New York University, New York City, NY Feb. 2026
- “Perspectives on Quantum Computing for Chemical Engineering” – Process Systems Engineering Seminar, Department of Chemical Engineering and Materials Science, University of Minnesota, Minneapolis, MN Nov. 2025
- “Perspectives on Quantum Computing for Chemical Engineering” – Process Systems Engineering Seminar, Massachusetts Institute of Technology, Cambridge, MA Video Nov. 2025
- “Reformulations and Decomposition for Quantum Discrete Optimization: Applications in Optimal Power Flow” – Operations Research Seminar, North Carolina State University, Raleigh, NC Mar. 2025
- “Reformulations and Decomposition for Quantum Discrete Optimization: Applications in Optimal Power Flow” – Industrial and Systems Engineering Research Seminar, Lehigh University, Bethlehem, PA Feb. 2025
- “Quantum–Classical Hybrid Methods: Applications in Optimization, Machine Learning, and Computational Chemistry” – Southeast Quantum Workshop, University of Tennessee, Knoxville, TN Nov. 2024
- “Discrete Nonlinear Optimization: Modeling and Solutions via Novel Hardware and Decomposition Algorithms” – Industrial and Systems Engineering Research Seminar, Lehigh University, Bethlehem, PA Feb. 2024
- “Modern Computational Approaches to Nonlinear Discrete Optimization and Applications in Process Systems Engineering” – Group for Applied Mathematical Modeling and Analytics, Industrial Engineering Department, University at Buffalo, Buffalo, NY Mar. 2021
- “Modern Computational Approaches to Nonlinear Discrete Optimization” – Process Systems Engineering Seminar, Department of Chemical and Biological Engineering, University of Wisconsin–Madison, Madison, WI Nov. 2020
- Internal seminars at Purdue (Birck Center for Nanotechnology, Apr. 2025; Midwest Quantum Collaboratory, Dec. 2023; Industrial Engineering Research Seminar, Sep. 2023)
- Guest lectures at Carnegie Mellon University (Programming Quantum Computers, Dec. 2020; Advanced PSE, Feb. 2020; Integer Programming, Feb. 2019)

Invited Seminars at US National Laboratories and Industry

- “Data Science in Chemical Engineering” – Corteva Agriscience, Indianapolis, IN May 2026
- “From Discrete Nonlinear Optimization to Hybrid Quantum Algorithms” – BASF Corporation, Houston, TX Apr. 2026
- “Quantum Optimization: The Intractable Decathlon Overview” – IBM Quantum Developer Conference, Atlanta, GA Video Nov. 2025
- “Quantum–Classical Hybrid Methods: Applications in Optimization, Machine Learning, and Computational Chemistry” – Center for Computing Research, Sandia National Laboratory, Albuquerque, NM Oct. 2024
- “Quantum–Classical Hybrid Methods: Applications in Optimization, Machine Learning, and Computational Chemistry” – Center for Nonlinear Studies, Los Alamos National Laboratory, Los Alamos, NM Oct. 2024
- “Quantum Computing for Optimization” – Modeling and Optimization Journal Club, Amazon, Seattle, WA Jan. 2021

International Invited Seminars

- “Perspectives on Quantum Computing for Chemical Engineering” – Department of Chemical Engineering, University College London, London, UK Jun. 2026

- “Quantum Optimization Benchmarking Library: The Intractable Decathlon” – Sargent Centre for Process Systems Engineering, Imperial College London, London, UK Jun. 2026
- “Perspectives on Quantum Computing for Chemical Engineering” – Department of Chemical and Process Engineering, University of Surrey, Guildford, UK Jun. 2026
- “Is this quantum computer useful? Let’s solve optimization with it! Evaluating the performance of quantum processing units at large width and depth” – CORMSIS Seminar, Centre for Operational Research, Management Sciences and Information Systems, University of Southampton, Southampton, UK Jun. 2026
- “Is this quantum computer useful? Let’s solve optimization with it! Evaluating the performance of quantum processing units at large width and depth” – Industrial Engineering Department Seminar, Bilkent University, Ankara, Türkiye Apr. 2026
- “Federated Learning for Process Systems Engineering: Foundations, Quantum Enhancements, and Applications” – 4th Process Systems Engineering State-of-the-Art Workshop, Istanbul, Türkiye Apr. 2026
- “Perspectives on Quantum Computing for Chemical Engineering” – Chemical Engineering Department Seminar, King Abdullah University of Science and Technology, Saudi Arabia Mar. 2026
- “Computing the Energy Transition: From Discrete Nonlinear Optimization to Hybrid Quantum Algorithms” – Mechanical Engineering Department Seminar, École Polytechnique Fédérale de Lausanne, Switzerland Mar. 2026
- “An Academic Perspective on International Collaboration for Quantum Computing” – New Paradigms Congress, Junta de Castilla y León, Spain Video Nov. 2025
- “Artificial Intelligence in Chemical Engineering” – Universidad de Costa Rica, San José, Costa Rica Aug. 2025
- “Perspectives on Quantum Computing for Chemical Engineering” – Universitat Politècnica de Catalunya, Barcelona, Spain Jun. 2025
- “Reformulations and Decomposition for Quantum Discrete Optimization: Applications in Optimal Power Flow” – Centre Automatique et Systèmes, Mines Paris–PSL, Paris, France Feb. 2025
- “Other Ideas to Leverage Quantum Computing for Discrete Optimization” – Optimization Workshop: Theory, Algorithms, and Applications, Universidad de los Andes, Bogotá, Colombia Dec. 2024
- “Perspectives on Quantum Computing for Chemical Engineering” – Universidad de Guanajuato, Mexico Oct. 2024
- “Discrete Nonlinear Optimization: Modeling and Solutions via Novel Hardware and Decomposition Algorithms” – Key Laboratory of Data Analytics and Optimization for Smart Industry, Northeastern University, Shenyang, China Apr. 2024
- “Quantum Computing for Process Systems Engineering” – 3rd Process Systems Engineering State-of-the-Art Workshop, Hangzhou, China Apr. 2024
- “Perspectives on Quantum Computing for Chemical Engineering” – Energy and Process Systems Engineering, ETH Zurich, Switzerland Mar. 2024
- “Discrete Nonlinear Optimization: Modeling and Solutions via Novel Hardware and Decomposition Algorithms” – Technology Innovation Institute, Abu Dhabi, United Arab Emirates Mar. 2024
- “Perspectives on Quantum Computing for Chemical Engineering” – Department of Chemical Engineering, Universidad Nacional de Colombia, Bogotá, Colombia Nov. 2023
- “Quantum and Quantum-Inspired Methods for Optimization” – 9th Arab-American Frontiers Symposium, NASEM, Doha, Qatar Oct. 2023
- “Discrete Nonlinear Optimization” – Learning from Both Sides: Linear and Nonlinear Mixed-Integer Optimization, Institut Mittag-Leffler, Stockholm, Sweden Jul. 2023
- “Discrete Nonlinear Optimization” – Power Systems Lab, ETH Zurich, Switzerland Feb. 2023
- “Perspectives on Quantum Computing for Chemical Engineering” – Sargent Centre for Process Systems Engineering, Imperial College London, UK Feb. 2023
- “Perspectives on Quantum Computing for Chemical Engineering” – Sustainable Process Systems Engineering Lab, ETH Zurich, Switzerland Dec. 2022
- “Perspectives on Quantum Computing for Chemical Engineering” – Department of Chemical Engineering, McMaster University, Hamilton, ON, Canada Oct. 2022

- “Quantum Computing and Modern Computational Optimization Approaches to PSE” – Laboratoire d’Informatique de Paris Nord, France Apr. 2022
- “Quantum Computing and Modern Computational Optimization Approaches to PSE” (in Spanish) – Universidad de los Andes, Bogotá, Colombia Jan. 2022
- “Modern Computational Approaches to Nonlinear Discrete Optimization and PSE” (in Spanish) – Universidad Nacional de Colombia, Bogotá, Colombia Dec. 2021
- Modeling and Optimization Group – PSR, Rio de Janeiro, Brazil Apr. 2021
- “Modern Computational Approaches to Nonlinear Discrete Optimization” – Real World Optimization Meeting, GOR and German Aerospace Center, Germany Mar. 2021
- “Modern Computational Approaches to Nonlinear Discrete Optimization” (in Spanish) – Department of Chemical Engineering, Universidad de Salamanca, Spain Jan. 2021
- “Quantum Integer Programming” – Department of Electrical Engineering, Indian Institute of Technology, Madras, India Oct. 2020
- “Incorporating Quadratic Approximations in the Outer Approximation Method for Convex MINLP” – Universidad Nacional del Litoral, Santa Fé, Argentina Aug. 2018
- “Incorporating Quadratic Approximations in the Outer Approximation Method for Convex MINLP” – Dagstuhl Seminar 18081, Dagstuhl, Germany Feb. 2018

Workshops, Panels, and Webinars

- Invited Workshop: “Hands-On Data Science for Chemical Engineering” – AIChE Spring Meeting, Houston, TX, USA Apr. 2026
- “AI and Quantum as the Frontiers in Process Systems Engineering” – CISTAR Annual Meeting, Chicago, IL, USA Apr. 2026
- Moderator and Panelist: “Quantum Computing for the Analytics Industry” – 2025 INFORMS Analytics+ Conference, Indianapolis, IN, USA Apr. 2025
- Invited Poster: “Quantum Artificial Intelligence” – 2nd Sustain Science in Latin America Symposium, NASEM, Lima, Peru Mar. 2025
- Invited Seminar: “Solving Unsolvable Problems: A Chemical Engineer’s Field Guide to Quantum Computing” – Process Development Division, AIChE Oct. 2025
- Panelist: “Artificial Intelligence in Chemical Engineering” – SOAIQ, Instituto Tecnológico de Monterrey, Mexico Oct. 2024
- “Advanced Optimization Methods and Hardware for Process Systems Engineering and Chemical Engineering” – CISTAR Biannual Meeting, University of New Mexico, Albuquerque, NM, USA Oct. 2024
- Webinar: “Quantum–Classical Hybrid Methods” – Qiskit Fall Fest IBM, Quantum Computing School in Spanish Sep. 2024
- Panelist: “Navigating the Intersection of Technology and Society” – Purdue Engineering Distinguished Lecture Series, West Lafayette, IN, USA Video Jan. 2024
- Panelist: “Quantum Computing” – 9th Arab-American Frontiers Symposium, NASEM, Doha, Qatar Oct. 2023
- Invited Lecture: “Quantum Computing Tutorial” – Machine Learning for Engineering, Universidad de los Andes, Bogotá, Colombia Nov. 2023
- Webinar: “Quantum and Quantum-Inspired Methods for Optimization” – Qiskit Fall Fest IBM, Quantum Computing School in Spanish Video Oct. 2023
- Webinar: “Perspectives on Quantum Computing for Chemical Engineering” – IChemE Webinar Series Video Oct. 2023
- Panelist: “Artificial Intelligence and Higher Education” – AI and Education Congress, Universidad Central, Bogotá, Colombia Oct. 2023

- Panelist: "Artificial Intelligence and Higher Education" – AI and Education Congress, Fundación Alberto Merani, Bogotá, Colombia Aug. 2023
- "Perspectives on Quantum Computing for Chemical Engineering: A Joint View from Academia and Industry" – CODES Research Group Meeting, University of West Virginia, Morgantown, WV, USA Feb. 2023
- "Exploiting and Benchmarking Ising Solvers" – Quantum Pittsburgh (QPitt) Meetup, Pittsburgh, PA, USA Oct. 2022
- "Introduction to Quantum Computing and Perspectives for Chemical Engineering" – Quantum Pittsburgh (QPitt) Meetup, Pittsburgh, PA, USA Jun. 2022
- "Modern Computational Approaches to Nonlinear Discrete Optimization and PSE" – Chemical Engineering Future Faculty Series Dec. 2020
- "Modern Computational Approaches to Nonlinear Discrete Optimization and PSE" – Discrete Optimization Talks (DOT) Video Dec. 2020
- "Quantum Computing for Discrete Nonlinear Optimization: Graver Augmented Multiseed Algorithm" – MINLP Virtual Workshop, Imperial College London, UK Video Jun. 2021

Invited Conference Presentations

1. **Bernal, D.E.** Tutorial: "Quantum computing for operations research: current state and perspectives," *International Federation of Operational Research Societies (IFORS) Conference 2026*.
2. **Bernal, D.E.**, Montanez-Barrera, J.A. "Is this quantum computer useful? Let's solve optimization with it! Evaluating the performance of quantum processing units at large width and depth," *International Federation of Operational Research Societies (IFORS) Conference 2026*.
3. **Bernal, D.E.**, Montanez-Barrera, J.A., Michielsen, K. "Is this quantum computer useful? Let's solve optimization with it! Evaluating the performance of quantum processing units at large width and depth," *2026 Society of Industrial and Applied Mathematics Optimization Meeting (SIAM OP26)*.
4. **Bernal, D.E.**. Panelist: "Panel on Quantum Computing and Operations Research," *2025 Institute For Operations Research and Management Science (INFORMS) Annual Meeting*.
5. **Bernal, D.E.**. "Continuous optimization for unconventional continuous computing: Accelerating Continuous Variable Coherent Ising Machines with momentum updates," *International Conference on Continuous Optimization (ICCOPT) 2025*.
6. **Bernal, D.E.**. "Perspectives of Quantum Computing in Chemical and Pharmaceutical Engineering," *Quantum ChemE 2025 (QChemE): Quantum Computing Applications in Chemical and Biochemical Engineering Workshop*.
7. **Bernal, D.E.**. "Reformulations and Decomposition for Quantum Discrete Optimization," *Workshop on Rivals to Quantum Computing*.
8. **Bernal, D.E.**. "Reformulations and Decomposition for Quantum Discrete Optimization: Applications in Optimal Power Flow," *2025 INFORMS Computing Society Meeting*.
9. Maciel Xavier, P., **Bernal, D.E.**. "Implementing Non-Standard Automatic QUBO Reformulation in JuMP," *2025 INFORMS Computing Society Meeting*.
10. Peng, Z., Cao, K., Furman, K., Li, C., Grossmann, I.E., **Bernal, D.E.**. "A Convexification-Based Outer-Approximation Method for Convex and Nonconvex MINLP," *2024 INFORMS Meeting*.
11. Peng, Z., Maciel Xavier, P., **Bernal, D.E.**. "Hybrid Quantum Branch-and-Bound Method for Quadratic Unconstrained Binary Optimization," *2024 INFORMS Meeting*.
12. **Bernal, D.E.**, Peng, Z., Furman, K., Li, C., Grossmann, I.E. "A Convexification-Based Outer-Approximation Method for Convex and Nonconvex MINLP," *25th International Symposium of Mathematical Programming (ISMP2024)*.
13. **Bernal, D.E.**, Peng, Z., Maciel Xavier, P. "Hybrid Quantum Branch-and-Bound Method for Quadratic Unconstrained Binary Optimization," *ISMP2024*.
14. Maciel Xavier, P., Ripper, P., Dias Garcia, J., Maculan, N., **Bernal, D.E.**. "QUBO.jl: A Tale of Implementation and Benchmarking of a Quantum Optimization Ecosystem in Julia," *ISMP2024*.

15. Bhatia, A.S., **Bernal, D.E.**. “Federated Hierarchical Tensor Networks: A Collaborative Quantum AI-Driven Framework for Healthcare,” *2024 INFORMS Optimization Society (OS) Meeting*.
16. Lubinski, T., Coffrin, C., McGeoch, C., Sathe, P., Apanavicius, J., **Bernal, D.E.**. “Optimization Applications as Quantum Performance Benchmarks,” *2024 INFORMS OS (IOS) Meeting*.
17. **Bernal, D.E.**, Brown, R., Sathe, P., Wudarski, P., Pavone, M., Rieffel, E., Venturelli, D. “Benchmarking the Operation of Quantum Heuristics and Ising Machines: Scoring Parameter Setting Strategies on Optimization Applications,” *2024 IOS Meeting*.
18. **Bernal, D.E.**. “Using Quantum and Physics-Inspired Methods for Constrained Optimization: Reformulations, Decomposition Algorithms, Software and Benchmarking,” *2023 INFORMS Meeting*.
19. Brown, R., **Bernal, D.E.**, Venturelli, D., Pavone, M. “Accelerating Coherent Continuous Variable Machines Using Momentum,” *2023 INFORMS Meeting*.
20. Sorourifar, F., Chamaki, D., Tubman, N., Paulson, J., **Bernal, D.E.**. “Specialized Gaussian Process Modifications for Shot-Efficient Quantum-Classical Optimization,” *2023 INFORMS Meeting*.
21. Peng, Z., Grossmann, I.E., **Bernal, D.E.**. “Mixed-Integer Nonlinear Decomposition Toolbox in Pyomo,” *2023 INFORMS Meeting*.
22. **Bernal, D.E.**, Brown, R.A., Venturelli, D., Pavone, M. “Hybrid Classical-Quantum Algorithms for Mixed-Integer Optimization,” *2023 SIAM Optimization Meeting (SIAM OP23)*.
23. **Bernal, D.E.**, Venturelli, D., Wudarski, F.A., Rieffel, E.G. “Benchmarking the Operation of Quantum Heuristics and Ising Machines: Scoring Parameter Setting Strategies on Real World Optimization Applications,” *2022 INFORMS Meeting*.
24. Brown, R.A., **Bernal, D.E.**, Venturelli, D., Pavone, M. “Copositive Optimization via Ising Solvers,” *2022 INFORMS Meeting*.
25. Rieffel, E., Kerger, P., **Bernal, D.E.**. “Quantum, Quantum-Classical Hybrid, and Distributed Quantum Algorithms for Problems in Operations Research,” *Workshop on Quantum Computing and Operations Research 2022*.
26. **Bernal, D.E.**. Plenary Moderator and Panelist: “Future of Quantum Computing in Optimization,” *2022 CORS/INFORMS International Conference*.
27. **Bernal, D.E.**, Grossmann, I.E. “Easily Solvable Convex Mixed-Integer Nonlinear Programs Derived from Generalized Disjunctive Programming Using Cones,” *2021 INFORMS Meeting*.
28. Quintero, R.A., **Bernal, D.E.**, Terlaki, T., Zuluaga, L. “Characterization of QUBO Reformulations for the Maximum k-Colorable Subgraph Problem,” *2021 INFORMS Meeting*.
29. **Bernal, D.E.**, Peng, Z., Kronqvist, J., Grossmann, I.E. “Regularization in Decomposition Methods for Global Optimization of Mixed-Integer Nonlinear Programming,” *31st European Conference on Operational Research (EURO) 2021*.
30. Li, C., Grossmann, I.E., **Bernal, D.E.**, Furman, K. “Sample Average Approximation for Stochastic Nonconvex Mixed Integer Nonlinear Programming via Outer Approximation,” *31st EURO 2021*.
31. Quintero, R.A., **Bernal, D.E.**, Terlaki, T., Zuluaga, L. “Characterization of QUBO Reformulations for the Maximum k-Colorable Subgraph Problem,” *31st EURO 2021*.
32. **Bernal, D.E.**, Kronqvist, J., Lundell, A., Grossmann, I.E. “A Review and Comparison of Solvers for Convex MINLP,” *2020 INFORMS Meeting*.
33. Li, C., **Bernal, D.E.**, Grossmann, I.E., Furman, K. “Sample Average Approximation for Stochastic Nonconvex Mixed Integer Nonlinear Programming via Outer Approximation,” *2020 INFORMS Meeting*.
34. **Bernal, D.E.**, Valentin, R., Chen, Q., Grossmann, I.E. “Mixed-Integer Nonlinear Decomposition Toolbox for Pyomo MindtPy,” *2019 INFORMS Meeting*.
35. Chen, Q., Valentin, R., Kale, S., Bates, J., **Bernal, D.E.**, Bynum, M.L., Siirola, J., Grossmann, I.E. “Advances in Pyomo.GDP: An Ecosystem for Nonlinear Disjunctive Programming Modeling and Optimization Development,” *2019 INFORMS Meeting*.
36. **Bernal, D.E.**, Gong, F., Chen, Q., Grossmann, I.E. “Mixed-Integer Nonlinear Decomposition Toolbox for Pyomo,” *2018 INFORMS Meeting*.

37. **Bernal, D.E.**, Kronqvist, J., Lundell, A., Westerlund, T., Grossmann, I.E. "A Center Cut Algorithm for Quickly Obtaining Feasible Solutions and Solving Convex Mixed-Integer Nonlinear Programs," *2018 INFORMS Meeting*.
38. Yang, H., **Bernal, D.E.**, Grossmann, I.E. "Integration of Crude-Oil Scheduling and Refinery Planning by Lagrangean Decomposition Approach," *2018 INFORMS Meeting*.

Contributed Conference Presentations

1. **Bernal, D.E.**. "Quantum Computing Applications in Chemical Engineering: Beginnings & Futures," *2025 American Institute of Chemical Engineers (AIChE) Annual Meeting*.
2. Park, A., **Bernal, D.E.**. "Quantum-Enhanced Federated Learning for Secure and Efficient Biomedical Image Classification," *2025 AIChE Annual Meeting*.
3. Ramesh, A., **Bernal, D.E.**. "Practical Framework for Assessing Performance of Quantum Simulation of Physical and Chemical Systems," *2025 AIChE Annual Meeting*.
4. Cabeza, A., Bardool, H., **Bernal, D.E.**. "Optimizing Membrane-Assisted Bio-Alcohol Dehydration Using Computational and Machine Learning Methods," *2025 AIChE Annual Meeting*.
5. Lee, A., **Bernal, D.E.**. "Integration of Derivative Free Optimization with Logic-Based Solutions of Generalized Disjunctive Programming," *2025 AIChE Annual Meeting*.
6. Tristán, C., **Bernal, D.E.**. "Optimizing Reverse Electrodialysis Design and Operation for Renewable Electricity Generation from Salinity Gradients," *2025 AIChE Annual Meeting*.
7. Maciel Xavier, P., **Bernal, D.E.**. "QUBO.Jl: A Julia Ecosystem to Equip Process Systems Engineering with Novel Optimization Paradigms," *2025 AIChE Annual Meeting*.
8. Cabeza, A., Orjuela, A., **Bernal, D.E.**. "Optimization Under Uncertainty for the Production of Tributyl Citrate through Simultaneous Acidification–Esterification Using Calcium Citrate," *2025 AIChE Annual Meeting*.
9. Bardool, H., **Bernal, D.E.**. "Computational Optimization and Machine Learning Modeling of Membrane-Assisted Bio-Methanol Dehydration," *2025 AIChE Midwest Regional Conference*.
10. **Bernal, D.E.**. "Generalized Disjunctive Programming: A Trip Down Memory Lane from Development, Extensions, and Future Perspectives," *2024 AIChE Annual Meeting*.
11. **Bernal, D.E.**. "Purdue Center for Operations and Optimization in Process Systems," *2024 AIChE Annual Meeting*.
12. Leal de Freitas, I., Peng, L., Dutta, S., **Bernal, D.E.**. "Quantum Federated Learning-Based Collaborative Manufacturing," *2024 AIChE Annual Meeting*.
13. Tristán, C., Fallanza, M., Ibañez, R., Grossmann, I.E., **Bernal, D.E.**. "Improving Sustainability in the Water Sector with Reverse Electrodialysis Optimization for Renewable Electricity Generation from Salinity Gradients," *2024 AIChE Annual Meeting*.
14. Peng, Z., Lee, A., **Bernal, D.E.**. "Addressing Discrete Dynamic Optimization via a Logic-Based Discrete-Steepest Descent Algorithm," *2024 AIChE Annual Meeting*.
15. Liñán, D.A., Lee, A., Ricardez-Sandoval, L., **Bernal, D.E.**. "A Combinatorial Benders Cuts Approach for Cost Minimization of Network Scheduling Problems with Mixed-Integer and Mixed-Integer Nonlinear Applications," *2024 AIChE Annual Meeting*.
16. Peng, Z., Maciel Xavier, P., **Bernal, D.E.**. "Hybrid Quantum Branch-and-Bound Method for Quadratic Unconstrained Binary Optimization," *2024 AIChE Annual Meeting*.
17. Sorourifar, F., Chamaki, D., Gustafson, E., Tubman, N., Paulson, J., **Bernal, D.E.**. "Bayesian Optimization-Aided Ground-State Molecular Calculation in Current Quantum Computers," *2024 AIChE Annual Meeting*.
18. **Bernal, D.E.**, Peng, Z., Maciel Xavier, P. "Hybrid Quantum Branch-and-Bound Method for Quadratic Unconstrained Binary Optimization," *Julia Mathematical Programming Developers Workshop 2024 (JuMP-dev 2024)*, Video.
19. Maciel Xavier, P., Ripper, P., Dias Garcia, J., Maculan, N., **Bernal, D.E.**. "QUBO.jl: A Tale of Implementation and Benchmarking of a Quantum Optimization Ecosystem in Julia," *JuMP-dev 2024*, Video.

20. Lee, A., Ovalle, D., Liñán, D.A., Ricardez-Sandoval, L., Gómez, J.M., Grossmann, I.E., **Bernal, D.E.**. "Logic-Based Discrete-Steepest Descent: A Solution Method for Process Synthesis Generalized Disjunctive Programs," *2024 AIChE Midwest Regional Conference*.
21. Tristán, C., Fallanza, M., Ibañez, R., **Bernal, D.E.**. "Optimizing Reverse Electrodialysis Process for Renewable Electricity Generation from Salinity Gradient," *2024 AIChE Midwest Regional Conference*.
22. **Bernal, D.E.**, Kerger, P., Rieffel, E. "Classical and Quantum Distributed Algorithms for the Survivable Network Design Problem," *2024 American Physical Society (APS) March Meeting*.
23. Gustafson, E., Tiihonen, J., Chamaki, D., Mullinax, W., **Bernal, D.E.**, Swaya, N., Maciejewski, F., Kim, J., Tubman, N., Krogel, J. "Surrogate Optimization for Quantum Circuits," *2024 APS March Meeting*.
24. Chamaki, D., Sorourifar, F., Velury, S., Hargus, C., Klymko, K., Hamilton, K., Hadfield, S., Mullinax, W., Paulson, J., **Bernal, D.E.**, Rotskoff, G., Tubman, N. "A Look at the Truths and Misconceptions of the Variational Quantum Eigensolver and the Implications of Overparameterization," *2024 APS March Meeting*.
25. Cabeza, A.F., Orjuela, A., **Bernal, D.E.**. "Tributyl Citrate Production from Calcium Citrate Salt: Reaction Kinetics and Process Simulation," *2023 American Institute of Chemical Engineers (AIChE) Meeting*.
26. Peng, Z., Grossmann, I.E., **Bernal, D.E.**. "Mixed-Integer Nonlinear Decomposition Toolbox in Pyomo," *2023 AIChE Meeting*.
27. **Bernal, D.E.**, Kerger, P., Rieffel, E.G. "Quantum Distributed Algorithms for Approximate Steiner Trees and Directed Minimum Spanning Trees," *2023 APS March Meeting*.
28. Sathe, P., Lubinski, T., Coffrin, C., Apanavicius, J., McGeoch, C., **Bernal, D.E.**. "Characteristics of Optimization Applications as Quantum Performance Benchmarks," *2023 APS March Meeting*.
29. Quintero, R.A., **Bernal, D.E.**, Terlaki, T., Zuluaga, L. "Characterization of QUBO Reformulations for the Maximum k-Colorable Subgraph Problem," *XXI Latin-Iberoamerican Conference on Operations Research (CLAIO) 2022*.
30. **Bernal, D.E.**. "Perspectives on Quantum Computing for Chemical Engineering: A Joint View from Academia and Industry," *2022 AIChE Meeting*.
31. **Bernal, D.E.**, Brown, R.A., Venturelli, D., Pavone, M. "Mixed-Binary Quadratic Programming via Convex Copositive Optimization and Ising Solvers," *7th International Conference on Continuous Optimization (ICCOPT) 2022*.
32. Brown, R.A., **Bernal, D.E.**, Sahasrabudhe, A., Loot, A., Venturelli, D., Pavone, M. "Copositive Optimization via Ising Solvers," *24th International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR) 2022*, Video.
33. **Bernal, D.E.**. "Perspectives on Quantum Computing for Chemical Engineering: A Joint View from Academia and Industry," *2022 Quantum Computing Applications in Chemical and Biochemical Engineering Workshop*.
    Prize for best talk at the workshop
34. **Bernal, D.E.**, Venturelli, D., Wudarski, F.A., Rieffel, E.G. "Benchmarking the Operation of Quantum Heuristics and Ising Machines: Scoring Parameter Setting Strategies on Real World Optimization Applications," *2022 APS March Meeting*.
35. **Bernal, D.E.**, Peng, Z., Kronqvist, J., Grossmann, I.E. "Alternative Regularization Schemes in Outer-Approximation Algorithms for Convex MINLP," *2021 AIChE Meeting*.
36. **Bernal, D.E.**, Ovalle, D., Liñán, D., Gómez, J.M., Ricardez-Sandoval, L., Grossmann, I.E. "Discrete-Steepest Descent: A Solution Method for Process Synthesis Generalized Disjunctive Programs," *2021 AIChE Meeting*.
37. Pedrozo, A., Rodriguez, S.B., **Bernal, D.E.**, Vechietti, A., Diaz, M.S., Grossmann, I.E. "Optimal Synthesis and Heat Integration Using Generalized Disjunctive Programming with Hybrid Models," *2021 AIChE Meeting*.
38. **Bernal, D.E.**, Peng, Z., Kronqvist, J., Grossmann, I.E. "Regularization in Decomposition Methods for Global Optimization of Mixed-Integer Nonlinear Programming," *2021 Society for Industrial and Applied Mathematics (SIAM) Conference on Optimization*.
39. **Bernal, D.E.** "Modern Computational Approaches to Nonlinear Discrete Optimization and Their Application to Process Systems Engineering," *2020 AIChE Meeting*.
    Meet the Faculty & Post-Doc Candidates Poster Session

40. **Bernal, D.E.**, Grossmann, I.E. "Use of Quantum Computing to Solve Optimization Problems in Process Systems Engineering," *2020 AIChE Meeting.*
Computing and Systems Technology Division Directors' Student Award Finalist
41. Li, C., **Bernal, D.E.**, Grossmann, I.E., Furman, K. "Sample Average Approximation for Stochastic Nonconvex Mixed Integer Nonlinear Programming via Outer Approximation," *2020 AIChE Meeting.*
42. Chen, Q., **Bernal, D.E.**, Johnson, E., Valentin, R., Kale, S., Bates, J., Siirola, J.D., Grossmann, I.E. "Pyomo.GDP: An Ecosystem for Logic-Based Modeling and Optimization Development," *2020 AIChE Meeting.*
43. Chen, Q., Liu Y., Seastream G., **Bernal, D.E.**, Siirola, J.D., Grossmann, I.E. "Pyosyn Graph: New Representation and Systematic Generation of Process Superstructures," *2020 AIChE Meeting.*
44. **Bernal, D.E.**, Booth, K.E.C., Dridi, R., Alghassi, H., Tayur, S., Venturelli, D. "Integer Programming Techniques for Minor-Embedding in Quantum Annealers," *Constraint Programming, Artificial Intelligence, Operations Research (CPAIOR) 2020.*
45. **Bernal, D.E.**, Grossmann, I.E. "Easily Solvable Convex Mixed-Integer Nonlinear Programs Derived from Generalized Disjunctive Programming Using Cones," *2019 AIChE Meeting.*
46. Chen, Q., Kale, S., Bates, J., Romeo, V., **Bernal, D.E.**, Bynum, M., Siirola, J.D., Grossmann, I.E. "Pyosyn: A Collaborative Ecosystem for Process Design Advancement," *2019 AIChE Meeting.*
47. **Bernal, D.E.**, Su, L., Tang, L., Grossmann, I.E. "Quadratic Cut Decomposition Method for Convex Mixed-Integer Nonlinear Programs," *2018 AIChE Meeting.*
48. Yang, H., **Bernal, D.E.**, Grossmann, I.E. "Integration of Crude-Oil Scheduling and Refinery Planning by Lagrangean Decomposition Approach," *2018 AIChE Meeting.*
Best submission in the Refining and Petrochemical Plant Modeling and Operations Improvement Session
49. Su, L., Tang, L., **Bernal, D.E.**, Grossmann, I.E., Wang, B. "Integrated Scheduling of On-Line Blending and Distribution of Oil Products in Refinery Operation," *13th International Symposium on Process Systems Engineering (PSE) 2018.*
50. Kronqvist, J., **Bernal, D.E.**, Grossmann, I.E. "A Level-Based Quadratic Outer-Approximation Algorithm for Convex MINLP," *2017 AIChE Meeting.*
51. **Bernal, D.E.**, Gomez, J.M. "Optimal Design and Control of a Catalytic Distillation Column: Case Study on Ethyl Tert-Butyl Ether (ETBE) Synthesis," *2016 AIChE Meeting.*
52. **Bernal, D.E.**, Vigerske, S., Trespalacios, F., Grossmann, I.E. "Feasibility Pump for Solving Convex MINLP Problems with DICOPT," *2016 AIChE Meeting.*

Theses

[T1] **D. E. Bernal**, "Modern Computational Approaches to Nonlinear Discrete Optimization and Applications in Process Systems Engineering.", Ph.D. Thesis available here, Ph.D. dissertation, Carnegie Mellon University, 2021.

[T2] **D. E. Bernal**, "Optimal design and control of a catalytic distillation column. Case study: Ethyl tert-butyl ether (ETBE) synthesis column.", M.Sc. Thesis available here, M.S. thesis, Universidad de los Andes, 2017.

[T3] **D. E. Bernal**, "Bounding the tangential velocities of Andromeda's satellite galaxies using nonlinear programming.", B.S. Thesis available (in Spanish) here, M.S. thesis, Universidad de los Andes, 2016.

[T4] **D. E. Bernal**, "Comparative study of the simulation methods of the extractive distillation system for the dehydration of ethanol using glycerol as a solvent.", B.S. Thesis available (in Spanish) here, M.S. thesis, Universidad de los Andes, 2014.

Memberships and Service

Leadership and Center Roles

- Co-director – Center for Operations and Optimization in Process Systems (COOPS), Davidson School of Chemical Engineering, Purdue University 2024–Current

- Inaugural Cohort – Purdue College of Engineering Emerging Research Leaders 2026–Current
- Artificial Intelligence Liaison – Davidson School of Chemical Engineering, Purdue University 2026–Current
- Program Chair – Gavriel Salvendy International Symposium on Frontiers in Industrial Engineering, Purdue Quantum AI (PQAI) Fall 2025
- Co-organizer – 2nd Colombian Academic Diaspora Symposium Summer 2025
- Co-chair – Quantum Computing Working Committee, INFORMS 2025–Current
- Represented the Purdue Quantum Science and Engineering Institute (PQSEI) at the launch of the International Year of Quantum at UNESCO, Paris, France Feb. 2025
- CACHE Ambassador – Computer Aids for Chemical Engineering 2025–Current

Institute and Center Affiliations (Purdue)

- Member – Center for Quantum Technologies (CQT) 2023–Current
- Member – Center of Innovative and Strategic Transformation of Alkane Resources (CISTAR) 2023–Current
- Member – Purdue Quantum Science and Engineering Institute (PQSEI) 2023–Current

University and Departmental Service (Purdue)

- Member – Ph.D. Recruitment Committee, Davidson School of Chemical Engineering Fall 2023–Current
- Faculty Advisor – American Institute of Chemical Engineers (AIChE) Student Chapter Fall 2023–Current
- Faculty Advisor – Colombian Student Association Spring 2024–Current
- Judge – 38 by 38 Award, Purdue College of Engineering Fall 2024, Fall 2025
- Marshall – Davidson School of Chemical Engineering Graduation Spring 2024

Editorial Activities

- Topic Editor – Frontiers in Computer Science, Experience with Quantum Annealing Computation 2022–2023
- Associate Editor – Frontiers in Chemical Engineering, Computational Methods in Chemical Engineering 2023

Conference and Workshop Organization

- Organizer – Quantum ChemE 2025 (QChemE): Quantum Computing Applications in Chemical and Biochemical Engineering Workshop, AIChE and DTU 2025
- Workshop Chair – Quantum Algorithms for Combinatorial Optimization, IEEE Quantum Week (QCE) 2025
- Workshop Chair – IBM Quantum Optimization Working Group, IEEE Quantum Week (QCE) 2025
- Organizer – Workshop on Quantum Computing and Artificial Intelligence for Chemical and Biochemical Engineering Applications, AIChE and DTU 2024
- Scientific Committee Member – Workshop on Quantum Computing for Chemical and Biochemical Engineering Applications, AIChE and DTU 2023
- Program Committee and Session Chair – JuMP-dev Annual Developer Meeting 2024
- Organizer – Mini-symposium on Quantum Computing in Optimization, SIAM OP Meeting 2023

Session Chairing and Program Roles (selected)

- Programming Coordinator – AIChE Annual Meeting, Applied Mathematics and Numerical Analysis (10D) 2027
- Track Chair – Computational Optimization, INFORMS Optimization Society Meeting 2024, 2026

- Session Chair – Quantum Computing and Operations Research (Quantum Optimization stream), IFORS Conference 2026
- Session Chair – Quantum Computing, INFORMS Optimization Society Meeting 2026
- Cluster Chair – Quantum Computing, INFORMS Computing Society Meeting 2025
- Session Chair – Quantum Computing and Operations Research, INFORMS Computing Society Meeting 2025
- Session Chair – Signs for a Potential Quantum Advantage in Optimization, APS Global Physics Summit 2026
- Session Chair – AIChE Annual Meeting, Quantum Computing Applications in Chemical Engineering: Beginnings & Futures 2025
- Session Co-chair – AIChE Annual Meeting, Applied Mathematics and Numerical Analysis (10D) 2025
- Session Chair – AIChE Midwest Regional Meeting, Process Systems Engineering 2025
- Session Co-chair – AIChE Annual Meeting, Advances in Process Design (10A) 2024
- Session Chair – AIChE Midwest Regional Meeting, Machine Learning and Optimizations 2024
- Session Chair (three sessions) – Quantum Computing in Continuous, Combinatorial, and Discrete Optimization, ICCOPT 2025
- Session Chair – Education in Process Design, Foundations of Computer Aided Process Design (FOCAPD) 2024
- Session Chair – Hybrid Quantum–Classical Methods for Optimization and Sampling, INFORMS Annual Meeting 2022
- Session Chair – Quantum Network Algorithms and Analysis, APS March Meeting 2023
- Sorter – Division of Quantum Information, APS March Meeting 2022
- Session Chair – Integer Programming and Pyomo sessions, INFORMS Annual Meeting 2019, 2021

Panels, Juries, and Other Service

- Panelist – The Revolution Before the Revolution: Technology Leadership in Quantum Computing, AIChE Annual Meeting 2024
- Panelist – Cafe con LatinX, AIChE Annual Meeting 2024
- Panelist – LatinXinChemE Annual Colloquium 2024
- Judge – Pritsker Doctoral Dissertation Award, Institute of Industrial and Systems Engineers 2025
- Panelist – Workshop on Quantum Computing and Operations Research, Fields Institute, Toronto 2022
- Jury – Best Poster, Chemical Engineering Division, LatinXChemE Forum 2021
- Jury – Best Undergraduate Graduation Project, Chemical Engineering, Universidad de los Andes 2021
- Moderator – Decomposition Techniques, MINLP Virtual Workshop 2021

Professional Society Memberships

- American Institute of Chemical Engineers (AIChE) Member since 2014
- Institute for Operations Research and the Management Sciences (INFORMS) Member since 2017
- Society for Industrial and Applied Mathematics (SIAM) Member since 2020
- American Physical Society (APS) Member since 2021
- Institute of Electrical and Electronics Engineers (IEEE) Member since 2023
- Computer Aids for Chemical Engineering (CACHE) Member since 2024
- Julia for Mathematical Programming (JuMP) – Developer Team Member since 2024

Peer Reviewer

- *Journals:* Mathematical Programming; SIAM Journal on Optimization; Chemical Engineering Journal; AIChE Journal; Chemical Engineering Science; Computers & Chemical Engineering; Computers & Operations Research; Industrial & Engineering Chemistry Research; Journal of Process Control; ACS Sustainable Chemistry & Engineering; Annals of Operations Research; EURO Journal on Computational Optimization; Nature Physics; Scientific Reports; European Journal of Operations Research (EJOR); Journal of Global Optimization; Journal of Optimization Theory and Applications; Optimization & Engineering; Optimization Letters; Digital Chemical Engineering; Current Opinion in Chemical Engineering; ACM Transactions on Quantum Computing; Quantum Information Processing; Quantum Machine Intelligence; Operational Research; Frontiers of Sustainability; International Journal of Energy Research; Acta Astronautica; IEEE Control Systems Letters (L-CSS).
- *Conferences:* International Conference on Machine Learning (ICML); Neural Information Processing Systems (NIPS); IEEE Conference on Decision and Control (CDC); IEEE International Conference on Quantum Computing and Engineering - Quantum Week (Quantum Machine Learning track 2025; Quantum Algorithms 2026); American Control Conference.
- *Grants:* Quantum Delta Netherlands National Grant.

Past Affiliations and Service (Carnegie Mellon University)

- Member – Pittsburgh Quantum Institute (PQI) 2019–Current
- Liaison – CMU Quantum Computing Group at PQI 2017–2021
- Organizer – Chemical Engineering LatinX and Hispanics Graduate Student Recruiting Session 2021
- Volunteer – SHPE Annual Meeting, CMU College of Engineering Graduate Student Recruiting 2020
- Conference Chair and Organizer – YinzOR Student Conference 2017

Languages

<table>
<tr><td rowspan="2"></td><td colspan="2">Comprehension</td><td colspan="2">Speaking</td><td rowspan="2">Written Expression</td></tr>
<tr><td>Listening</td><td>Reading</td><td>Oral Interaction</td><td>Oral Production</td></tr>
<tr><td>Spanish</td><td colspan="5">Native Speaker</td></tr>
<tr><td rowspan="2">English</td><td>C2</td><td>C2</td><td>C1</td><td>C1</td><td>C1</td></tr>
<tr><td colspan="5">TOEFL: 115/120</td></tr>
<tr><td rowspan="2">German</td><td>C1</td><td>C1</td><td>C1</td><td>B2</td><td>B2</td></tr>
<tr><td colspan="5">TestDaF: 4.5/5</td></tr>
<tr><td rowspan="2">French</td><td>A2</td><td>A2</td><td>A2</td><td>A2</td><td>A1</td></tr>
<tr><td colspan="5">DELF A2: 74.5/100</td></tr>
</table>

Level A1/A2: Basic User — B1/B2: Independent User — C1/C2: Proficient User
(Common European Framework of Reference for Languages)